

Group Project: MicroOdyssey — the microbial journey of early life

Description of challenge:

A new ETH research team is on a mission to read time in the gut. Every baby's microbiome matures on its own schedule during the first two years of life, and is convinced that this developmental journey is written into the metagenome – if you know how to read it. You have been handed shotgun metagenomic sequencing reads from one study, including 234 serial stool samples collected from 55 infants between birth and 24 months of age, paired with metadata capturing each infant's age, delivery mode, feeding and more. Your mission, titled "MicroOdyssey", is to chart how microbial diversity and composition change as infants grow, and to work out how much of that trajectory is shared across babies and how much is unique to each one.

With MicroOdyssey, the team hopes to build the first microbial "odyssey" of infancy. What hidden trajectories guide the microbial journey of early life? Your task is to find this out.

Some specific questions*:

- a) How do the stool samples differ in microbial diversity according to age and between the different infants?

Hint: profile the shotgun reads taxonomically with a read-based classifier (e.g. Kraken2 with Bracken), then compute alpha- and beta-diversity metrics on the resulting profiles.

- b) Each infant is sampled repeatedly across the first two years of life. How does within-sample (alpha) diversity change with age, and do the trajectories differ from infant to infant? When you look at between-sample (beta) diversity, is the structure driven more strongly by age or by infant identity (host_id)?
- c) The metadata also records delivery mode (vaginal vs C-section), feeding mode (diet_milk) and infant sex. Do any of these factors relate to microbial diversity or to how quickly the microbiome matures with age? Are there other factors in the metadata that seem to influence the developing microbiome?
- d) On top of diversity, microbial composition is also very important in early life. Investigate compositional changes (e.g. phylum, family, genus) in the samples along age and between the different infants. Which taxa mark the transition from the newborn gut to the older-infant gut, and are there compositional markers that distinguish individual infants' trajectories?

Information on provided datasets:

Reads: The .qza artifact contains the short shotgun metagenomic sequencing reads you will use (paired-end, sequenced on an Illumina NovaSeq 6000). Due to the time constraints of the course, the reads have already been pre-processed with adapter trimming, quality control and host (human) read removal, and are ready for you to use. You will analyse only the reads – no metagenome-assembled genomes (MAGs) or contigs are provided or required to construct.

Sample metadata: The metadata table contains additional information on each collected sample. Each row represents a unique sample that can be identified with its unique id. Columns include the infant identifier (host_id), age (host_age_days, host_age_months), delivery mode, feeding mode (diet_milk), sex and sequencing depth, among others.

Where to get the data:

Reads: Please e-mail the contact TA to get the metagenomics reads data.

Sample metadata: <https://polybox.ethz.ch/index.php/s/iodBC3wfxeHMSDZ>

Questions? Your course instructors are here to help. Meet with instructors during in-class group work sessions or ask questions via email.

Contact TA: Fannie Kerff, fannie.kerff@hest.ethz.ch

* Note on questions: This is not a list of the only questions to answer about this dataset. All group semester projects are expected to incorporate most analysis topics covered in the course when possible (see Group Semester Projects assignment document). This is merely a list of specific questions that should be answered / attempted in addition to the “standard” analyses covered. Not everything will work or give a clear result! But that does not mean that you should not try. Your goal as scientists is to figure out why.